Tiago Ferrão-Lindqvist

Test & Validation Engineer | Python and SQL Data Tooling

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PROFESSIONAL SUMMARY

Mechanical engineer with 12 years of experience in automotive and industrial product design, testing and validation, transitioning into data engineering. Built scheduled Python data-processing and reporting tooling used by four other engineers, integrating test files with SQL Server and Power BI. Brings practical experience in specification-based data checks, reusable internal tools and engineering stakeholder needs. Portuguese native speaker with fluent English.

TECHNICAL SKILLS

- Python: pandas, NumPy, Matplotlib, npTDMS; file parsing, specification checks and report generation.
- SQL and BI: SQL Server queries for trend analysis, summary-result writes to an IT-created staging table, Power BI; SQLite in a personal project.
- Automation: Windows Task Scheduler for nightly script execution.
- Development tools: Git in personal repositories since 2025; Flask and tutorial-based Docker Compose use in a local learning project.
- Engineering: MATLAB/Simulink, LabVIEW, ANSYS Mechanical, SolidWorks, design of experiments, DFMEA and 8D problem solving.

PROFESSIONAL EXPERIENCE

Test & Validation Engineer | Corvane Automotive Components

Setúbal, Portugal | Mar 2019 - Present

- Built Python scripts to parse TDMS and CSV output from six test rigs, processing about 2 GB per week with pandas, NumPy and npTDMS.
- Automated checks of every channel against test specifications and generated standard report figures with Matplotlib, reducing report preparation from about three days to half a day per test.
- Run the scripts nightly on a lab PC through Windows Task Scheduler and write summary results to a SQL Server staging table created by IT; four other test engineers use the internal tool.
- Query the plant's SQL Server test-results database to analyse trends across three product generations.
- Plan and run durability, vibration and climatic tests on seat-adjuster mechanisms and prepare reports for OEM customers.
- Correlate test results with FEA and lead 8D root-cause analysis; a proposed adjuster rail bracket redesign reduced warranty returns for that part by 40%.
- Trained three apprentices in rig safety and test setup between 2022 and 2024.

Mechanical Design Engineer | Halvik Pumps

Porto, Portugal | Sep 2014 - Feb 2019

- Designed pump casings and impellers in SolidWorks and owned DFMEA for two product lines.
- Wrote a LabVIEW acquisition program for the performance test bench, reducing test setup time from about two hours to 30 minutes.
- Performed structural FEA in ANSYS; a casing-family change from grey cast iron to ductile iron saved 12% in material cost.

SELECTED PROJECTS

Test results dashboard | Corvane Automotive Components | 2023

- Built a Power BI dashboard on the SQL Server staging table showing pass/fail trends and cycles to failure by product generation; used by about 15 validation and design engineers.

Bike-sharing trips API | Personal learning project | 2025

- Built a small Flask API with three read-only endpoints over a public bike-sharing dataset loaded into SQLite.
- Runs locally on a laptop; followed a tutorial to run it with Docker Compose. Technologies: Python, Flask, SQLite, Docker Compose and Git.

EDUCATION AND PROFESSIONAL DEVELOPMENT

MSc Mechanical Engineering | Sado Valley Polytechnic | 2014

Thesis: Fatigue life prediction of welded brackets under variable-amplitude loading

Online data engineering course | In progress

Completed six of nine modules; no certificate awarded yet.

LANGUAGES

Portuguese: native | English: fluent | Spanish: good